

## **Challenge 9 – Smart Industrial Pollution Monitoring for Golden Corridor (Vapi–Ankleswar)**

**Domain: Environmental Sustainability**

### **The Challenge**

The industrial belt along Vapi, Ankleshwar, and Vatva faces significant air and water pollution from chemical and manufacturing units, with limited real-time monitoring and slow regulatory response to violations.

### **Objective**

Build Agentic AI solutions that continuously monitor industrial emissions and effluents, detect violations, and trigger timely regulatory and community alerts.

### **Suggested AI Agents**

- Emission & Effluent Monitoring Agent
- Violation Detection & Compliance Agent
- Regulatory Alert & Escalation Agent
- Public Health Risk Assessment Agent
- Industrial Pollution Dashboard Agent

**Technology to be used – IBM Bob + IBM Granite LLM + IBM cloud**

Create the agent using IBM Granite LLM using following credentials.

project id : - 29f2b75e-ec3e-463c-b6de-aefa520a2d81

api key : \_DYH5uur3l01T5cJnDFocOIDJArrajmrBTpa-yVp1jHR

url : <https://api.au-syd.watson-orchestrate.cloud.ibm.com/instances/012a2164-ac43-4d87-a90b-39ed1e476339>

Crete user interface using react JS.

Install necessary software from your end.

Create the necessary exe files(.bat files) so that user will click on .bat file and run the project from his end.

Run the project from your end.

Create Public URL.

Rename the .env-example to .env and add credentials.

Create the agent using IBM Granite 4-h small using following credentials.